Lesson plan · for the teacher

Decisions: if, elif and else

Programming · Level 2 · Lesson 2

Time
40 min

Materials
Board and chalk; Cards with made-up marks, including 49, 50, 69 and 70; Your phone with this lesson downloaded

Learning goals

- Compare values with ==, <, >= and the others.
- Write if, elif and else with colons and indents.
- Test a marking rule at its boundaries.

Activity: Find your corner

1. Warm-up (5 min): Call out conditions: “Stand up if you wear black shoes”, “...if your birthday is in May or June”. Agree that a condition is either true or false.
2. Teach (10 min): Write the grading rule on the board as if, elif and else. Show the colon and the indent. Ask: why must “70 or more” come before “50 or more”?
3. Sort (10 min): Give each pupil a mark card, including 49, 50, 69 and 70. Label three corners A, B and C. Pupils go to their corner; the class checks each one with the rule.
4. Run (10 min): Type the marks that caused arguments into the program on your phone and read out the grades. Did the program agree with the class?
5. Wrap-up (5 min): Ask which marks were hardest, and why boundaries cause bugs. Then give the exit ticket.

One phone is enough: you run the program, and the pupils are the data. Download the lesson the day before so Python works offline.

Discussion

- Why is 50 the mark most likely to be graded wrong?
- What goes wrong if the B test comes before the A test?
- Which other school rule could be an if statement?

Exit ticket

- Write the marks you would test for “40 or more passes”, and why.

Try it at school tomorrow
Write your school’s grading rule on the board as if, elif and else. Give pupils five marks, including 49 and 50, and ask which grade each one gets. Then run them on your phone: did the program agree?

Practice questions · for you

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1 **A pupil writes this line with one equals sign. What happens when the program runs?**

```
if mark = 50:
    print("Pass")
```

◯ It sets mark to 50, then prints Pass ◯ Python stops with a SyntaxError ◯ It works the same as ==

2 **In an if … elif … else, Python runs every branch whose condition is true.**

◯ True ◯ False

3 **Put the lines in order: the program asks for a mark and prints Pass for 50 or more, and Try again for anything less. One line doesn't belong.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `if mark >= 50:`

☐ `print("Pass")`

☐ `print("Try again")`

☐ `if mark > 50:`

☐ `mark = int(input("Mark? "))`

☐ `else:`

4 **The rule is "70 or more is A, 50 or more is B, else C", but a pupil with exactly 50 gets C. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  mark = int(input("Mark? "))
2  if mark >= 70:
3      print("A")
4  elif mark > 50:
5      print("B")
6  else:
7      print("C")
```

5 **Match each comparison to its meaning.**

mark == 50	• • • •	mark is anything but 50
mark != 50	• • • •	mark is exactly 50
mark >= 50	• • • •	below 50
mark < 50	• • • •	50 or more

✎ **Which rule in your school could a program apply for you: grades, fees, attendance? Write it as if, elif and else, and list the boundary values you would test.**

Check your answers

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1 **Python stops with a SyntaxError**

Inside an if, a single = is a syntax error: = stores, == compares.

2 **False**

Only the first true branch runs; the rest are skipped.

3 **Top to bottom: 2, 3, 5, X, 1, 4**

```
mark = int(input("Mark? "))
if mark >= 50:
    print("Pass")
else:
    print("Try again")
```

The mark is read first, then tested; print lines are indented under if and else.

4 **Line 4: `elif mark >= 50:`**

> leaves out the boundary: 50 is not greater than 50, so it falls through to else.

5 **mark == 50 — mark is exactly 50**
mark != 50 — mark is anything but 50
mark >= 50 — 50 or more
mark < 50 — below 50

Two equals signs compare; != means "not equal".